

RESEARCH ARTICLE

Heterogeneity and Heteroskedasticity in Endogenous Switching Models: Estimating the Effects of Physician Advice on Calorie Consumption

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ABSTRACT

We describe two-step control function estimation procedures for estimating average treatment effects in constant coefficient endogenous switching models with a binary treatment. We allow for additional heterogeneity by allowing for heteroskedasticity in the latent error in the treatment assignment equation. We apply our estimation procedures to evaluate the causal effect of physician advice on calorie consumption using National Health and Nutritional Examination data (2007–2016) and find that the effect of physician advice to lose weight on average calorie consumption roughly doubles compared to the traditional methods. Our findings recognize physician advice as a relevant low-cost intervention in the campaign against rising obesity.

JEL Classification: C21, C24

1 | Introduction

We propose parametric and semiparametric two-step control function estimation procedures for Endogenous Switching Models (ESMs) with a heteroskedastic binary treatment variable for cross section data. A typical ESM regression model with an observed outcome variable y_i and observed covariates x_i is given as $y_i = (1 - w_i)(x_i\beta_0 + u_{i0}) + w_i(x_i\beta_1 + u_{i1})$ where w_i is the switching variable modeled as a binary endogenous variable $w_i = \mathbb{1}[z_i\gamma + v_i \geq 0]$ with u_{i0}, u_{i1}, v_i denoting the unobservables. The traditional ESM postulates the existence of two “Regimes: Regime 0 and Regime 1” and w_i captures what “Regime” an individual belongs to. The two expressions $x_i\beta_0 + u_{i0}$ and $x_i\beta_1 + u_{i1}$ depict the regressions of the output in the two regimes respectively. The endogeneity issue arises because the individual endogenously decides what regime to choose. For example, w_i could denote the endogenous decision to adopt a new agricultural technology and

the output variable is the crop yield corresponding to the value of w_i . An individual can only belong to one “Regime” (either adopt or not adopt), and we observe only one outcome variable corresponding to that “Regime.”

In this paper, we allow v_i to have conditional multiplicative heteroskedasticity that is an unknown function of observable covariates. To control for endogenous treatment assignment, we use conditional linear projections (CLPs) to obtain the correction functions that are allowed to be functions of observable covariates. There are several examples in the empirical research that highlight the importance of allowing for a heteroskedastic treatment assignment model. For example, Abdel-Rahman et al. (2021) show that it is crucial to incorporate heteroskedasticity in estimating the effect of catastrophic out-of-pocket health expenditure because specific covariates significantly influence the probability of incurring catastrophic

payments through health payment variance. Mu and Zhang (2018) use heteroskedastic binary choice models to study the intergenerational transmission of smoking habits in British households.

In this paper, we explore the impact of physician advice to lose weight on calorie consumption through our model and proposed estimation procedures. Research has shown that while there are positive impacts of physician advice to lose weight, there are substantial disparities related to observables and unobservables patient characteristics in who receives weight-loss advice from physicians (Lorts and Ohri-Vachaspati 2016). Specifically, the probability of getting a physician advice on weight related issues is often correlated with several observed sociodemographic factors such as income and also unobserved factors such as implicit bias. We allow for a multiplicative conditional heteroskedastic treatment assignment model that helps to capture this heterogeneity in the propensity to get treatment in a flexible manner. We find substantial differences in the impact of physician advice on calorie consumption when we allow for a heteroskedastic treatment model compared to standard estimation procedures such as ordinary least squares (OLS), two-stage least squares (2SLS), and even ESMs with a homoskedastic treatment model.

1.1 | Related Literature

ESMs provide a framework to estimate heterogeneous treatment effects when the treatment is defined to be a binary variable (such as union status or some government intervention) and is suspected to be correlated with the unobservables even after conditioning on observables (Wooldridge 2010). Many questions in economics, such as inter-industry and inter-sector wage differentials (Adamchik and Bedi 2000; Sakellariou 2012), wage differentials between union and nonunion members (Lee 1978; Nahm et al. 2017), housing demand (Murtazashvili and Wooldridge 2016), and agricultural economics and health economics (Abdulai and Huffman 2014; Loureiro and Nayga 2006; Mojo et al. 2017; Wilde and Ranney 2000) have been studied under the framework of ESMs.

Parametric estimation procedures for ESMs are based on the distributional assumptions between the unobservables. Maximum likelihood methods typically assume the distribution of the unobservables to be a trivariate normal distribution. We also have two-step estimation procedures that control for endogeneity through correction functions that embody the endogenous relationship between the unobservables. These two-step estimators are obtained under an array of distributional assumptions with varying degrees of restrictiveness that is reflected in the specifications of the correction functions. For example, when the unobservables are assumed to have a trivariate normal distribution with constant variances, nonzero covariances between (u_{i0}, v_i) and (u_{i1}, v_i) capture the endogeneity. Under normality, the conditional expectations of the unobservables $E[u_{ig}|v_i]$ are linear in v_i for $g = 0, 1$ and are estimated in the first step.¹ Alternatively, one can use linear projections to model the conditional expectations if the normality assumption seems too restrictive, as is done in the estimation procedures described in Murtazashvili and Wooldridge (2016). Other estima-

tion procedures such as Choi and Min (2009); Lee (1984) accommodate deviations from normality. Finally, one can also obtain semiparametric two-step estimation procedures that relax distributional assumptions either one or both the steps of estimation.

Two-step control function estimating procedure often have the first step of estimating the model for the treatment assignment. When the treatment variable is binary, this entails estimating a probit model through maximum likelihood estimation (MLE). If there is heteroskedasticity in the treatment assignment model, ignoring it in the first step estimation can give misleading results due to misspecification. As we mentioned earlier, many program evaluation models suggest presence of heteroskedasticity in the treatment assignment. Despite this, current literature on ESM and on two-step estimation procedures with heteroskedastic error in the first step is limited.

In this paper, we allow the unobservables in the model for the treatment variable to have conditional multiplicative heteroskedasticity. In other words, we allow $v_i \sim Normal(0, \sigma_v^2(z_i))$ where $\sigma_v^2(z_i)$ is a function of the observed covariates, allowing for the joint distribution of our unobservables to depend on the observed covariates.

We apply our model and estimation procedures to study the impact of the health care intervention of physician advice to lose weight on calorie consumption. We find that allowing for a heteroskedastic error structure in the model for who receives physician advice to lose weight captures the contribution of an individual's observed and unobserved socio-economic factors (such as income, weight, bias etc) in a flexible way that is captured by $\sigma_v^2(z_i)$. We use Conditional Linear Projections (CLPs) to obtain the correction functions that capture heterogeneity in treatment assignment which is allowed to be a flexible function of the covariates. Most recently, Carlson and Joshi (2024) used CLPs to allow for heterogeneous coefficients in panel data sample selection models. Our paper provides an extension of their methodology and applies CLPs to develop estimators for ESMs that allow researchers to incorporate increased heterogeneity in treatment assignment.

1.2 | Organization of the Paper

Section 2 describes the model specification, assumptions, and estimation equation for the constant coefficient endogenous switching models. Section 3 describes the two-step parametric and sieve estimation procedures. It also provides a brief discussion of key themes to consider for the identification of the parameters. In Section 4, we illustrate small sample properties of our proposed estimators through limited Monte Carlo simulation experiments. The simulations illustrate that our proposed estimators perform superior to common estimators in the presence of heteroskedasticity in the treatment assignment model. Finally, in Section 5, the proposed estimators are applied to study the impact of physician advice to lose weight on calorie consumption. We find that incorporating heteroskedasticity in the model for receiving physician advice provides substantial evidence that supports physician advice as a low-cost intervention to control obesity.

2 | Model and Estimating Equation

For a random sample of N observations, we postulate the existence of two potential outcomes, denoted by $\{y_i^0, y_i^1\}$, where y_i^0 is the outcome in the untreated (or control) state and y_i^1 is the outcome in the treated state. One can also interpret the potential outcomes as the outcomes obtained in each “Regime” with “Regime = 0” denoting the untreated state and “Regime = 1” denoting the treated state. The treatment status of an individual i is denoted by a binary variable w_i that takes the value 1 if she is treated and 0 otherwise. Our first assumption states that these counterfactual outcomes are linear in the parameters²:

Assumption 2.1. For each $i = 1, 2, \dots, N$,

$$\begin{aligned} y_i^0 &= \alpha^0 + x_{i1}\beta^0 + u_i^0 \\ y_i^1 &= \alpha^1 + x_{i1}\beta^1 + u_i^1 \end{aligned}$$

where x_{i1} is a $1 \times K_1$ vector of exogenous explanatory variables (excluding a constant) that allows interactions and functional forms such as log, squares that are linear in parameters. $\{u_i^0, u_i^1\}$ are the unobservables. $\square$

The observed outcome y_i is obtained as

$$\begin{aligned} y_i &= (1 - w_i)y_i^0 + w_i y_i^1 = (1 - w_i)(\alpha^0 \\ &+ x_{i1}\beta^0 + u_i^0) + w_i(\alpha^1 + x_{i1}\beta^1 + u_i^1) \end{aligned} \quad (2.1)$$

which gives the switching regression model. We model our treatment variable w_i as a binary random variable with conditional multiplicative heteroskedasticity:

Assumption 2.2. For each $i = 1, 2, \dots, N$

$$w_i = \mathbb{1}[z_i\gamma + v_i \geq 0] \quad (2.2)$$

where the errors are multiplicative heteroskedastic. The conditional distribution of $v_i|z_i$ is specified as

$$v_i|z_i \sim \text{Normal}(0, \sigma_v^2(z_i)) \quad (2.3)$$

and $\sigma_v(z_i)$ is a function of all exogenous explanatory variables. $\square$

Next, we assume that we have a set of exogenous variables $z_i \equiv \{x_{i1}, x_{i2}\}$ where x_{i2} is a $1 \times K_2$ vector of observed covariates (also called exogenous instruments) that may influence the treatment mechanism but not the outcome equation. The following assumption formalizes the nature of the exogeneity assumption³:

Assumption 2.3. For each $i = 1, 2, \dots, N$ we assume that (a) $E[u_i^0|z_i] = E[u_i^1|z_i] = 0$ and (b) $K_2 \geq 1$ $\square$

2.1 | Average Treatment Effect (ATE)

The Average Treatment Effect is defined as $\tau_{ate} = E[y_i^1] - E[y_i^0] \equiv \mu_1 - \mu_0$ and under Assumption 2.3, $\tau_{ate} = (\alpha^1 - \alpha^0) + E[x_1](\beta^1 - \beta^0)$. In the switching regression model of

Equation 2.1, τ_{ate} is captured by the coefficient on w_i . To see this, note that

$$y_i = \alpha^0 + x_{i1}\beta^0 + w_i[\alpha^1 - \alpha^0] + w_i x_{i1}(\beta^1 - \beta^0) + u_i^0 + w_i(u_i^1 - u_i^0).$$

Substitute $(\alpha^1 - \alpha^0) = \tau_{ate} - E[x_1](\beta^1 - \beta^0)$ to get

$$\begin{aligned} y_i &= \alpha^0 + x_{i1}\beta^0 + w_i(\mu_1 - \mu_0) + w_i[x_{i1} \\ &- E[x_1]](\beta^1 - \beta^0) + u_i^0 + w_i(u_i^1 - u_i^0). \end{aligned}$$

Denoting $\alpha \equiv \alpha^0$, $\beta_0 \equiv \beta^0$, $\beta_1 \equiv (\beta^1 - \beta^0)$, $u_{i0} \equiv u_i^0$ and $u_{i1} \equiv (u_i^1 - u_i^0)$, we get

$$y_i = \alpha + w_i\tau + x_{i1}\beta_0 + w_i(x_{i1} - E[x_1])\beta_1 + u_{i0} + w_i u_{i1}. \quad (2.4)$$

In this specification, we get $\tau \equiv \tau_{ate}$ directly as the coefficient on w_i .

2.2 | Estimating Equation

To obtain the estimation equation, start with

$$E[y_i|z_i, w_i] = E[E[y_i|z_i, v_i]|z_i, w_i] \quad (2.5)$$

$$\begin{aligned} &= \alpha + w_i\tau + x_{i1}\beta_0 + w_i(x_{i1} - E[x_1])\beta_1 \\ &+ E[E[u_{i0}|z_i, v_i]|z_i, w_i] + w_i E[E[u_{i1}|z_i, v_i]|z_i, w_i] \end{aligned} \quad (2.6)$$

where the first equality follows from w_i being a function of (z_i, v_i) . To account for the bias due to the endogenous switching, we need expressions for $E[u_{i0}|z_i, v_i]$ and $E[u_{i1}|z_i, v_i]$. We want the option of having the conditional heteroskedasticity to be of an unknown form. Thus, we impose linearity only with respect to v_i , leaving the functional form for $E[u_{i0}|z_i, v_i]$ and $E[u_{i1}|z_i, v_i]$ unspecified in z_i . This is done by using conditional linear projections (or predictions).

Denoted by $\mathbb{L}[u_{ij}|v_i; z_i]$, conditional linear projection (CLP) is defined as the linear projection of u_{ij} on v_i conditional on z_i , for $j = \{0, 1\}$. The theory of CLPs was first developed in Hansen and Richard (1987) in which they essentially extended the conventional Hilbert Space analysis to the conditional framework. Wooldridge (1999) uses the concept of CLPs to develop the orthogonality conditions to obtain a distribution-free estimation of nonlinear panel data models. More recently, Carlson and Joshi (2024) use CLPs to allow for heterogeneous coefficients in sample selection models for panel data. We use CLPs to model the endogenous switching as

Assumption 2.4. The conditional expectations are linear in v_i conditional on z_i :

$$E[u_{i0}|v_i, z_i] = \mathbb{L}[u_{i0}|v_i; z_i] \equiv \left(\frac{\sigma_0(z_i)}{\sigma_v^2(z_i)} \right) v_i \quad (2.7)$$

$$E[u_{i1}|v_i, z_i] = \mathbb{L}[u_{i1}|v_i; z_i] \equiv \left(\frac{\sigma_1(z_i)}{\sigma_v^2(z_i)} \right) v_i \quad (2.8)$$

where $\sigma_j(z_i) \equiv \text{Cov}(u_{ij}, v_i|z_i)$, $j = \{0, 1\}$ denote the conditional covariances between the error terms in the outcome equation and

the error term in the reduced form equation for the treatment variable. $\square$

Assumption 2.4 imposes linearity of $E[u_{ij}|v_i, z_i], j = \{0, 1\}$ but only with respect to v_i . Because we are using linear projections conditional on z_i , we still allow $E[u_{ij}|v_i, z_i]$ to be a flexible nonlinear function of z_i . This is the key difference with traditional control function approaches to endogenous switching models, that typically impose $E[u_{ij}|v_i, z_i] = E[u_{ij}|v_i] = \rho_j v_i, j = \{0, 1\}$, where ρ_0, ρ_1 are constants.

Combined with the flexible probit model for w_i , Assumption 2.4 thus allows considerable heterogeneity.⁴ Next, substitute (2.7) and (2.8) into Equation (2.5) to get

$$\begin{aligned} E(y_i|z_i, v_i) &= \alpha + w_i\tau + x_{i1}\beta_0 + w_i(x_{i1} - E[x_1])\beta_1 \\ &+ \left(\frac{\sigma_0(z_i)}{\sigma_v^2(z_i)}\right)v_i + w_i\left(\frac{\sigma_1(z_i)}{\sigma_v^2(z_i)}\right)v_i. \end{aligned} \quad (2.9)$$

Next, we have $E[v_i|z_i, w_i]$ as the generalized error from the model for w_i as

$$\begin{aligned} E(v_i|z_i, w_i) &= \sigma_v(z_i) \left[w_i \lambda\left(\frac{z_i\gamma}{\sigma_v(z_i)}\right) - (1 - w_i) \lambda\left(-\frac{z_i\gamma}{\sigma_v(z_i)}\right) \right] \\ &\equiv \sigma_v(z_i) h(w_i, z_i) \end{aligned}$$

where $\lambda(\cdot)$ is the Inverse Mills Ratio of the standard normal distribution with $\lambda\left(\frac{z_i\gamma}{\sigma_v(z_i)}\right) \equiv \frac{\phi\left(\frac{z_i\gamma}{\sigma_v(z_i)}\right)}{\Phi\left(\frac{z_i\gamma}{\sigma_v(z_i)}\right)}$ and $\lambda\left(-\frac{z_i\gamma}{\sigma_v(z_i)}\right) \equiv \frac{\phi\left(-\frac{z_i\gamma}{\sigma_v(z_i)}\right)}{\Phi\left(-\frac{z_i\gamma}{\sigma_v(z_i)}\right)}$. Finally, letting $f_{v_0}(z_i) \equiv \frac{\sigma_0(z_i)}{\sigma_v(z_i)}$ and $f_{v_1}(z_i) \equiv \frac{\sigma_1(z_i)}{\sigma_v(z_i)}$, we get the estimating equation as

$$\begin{aligned} y_i &= \alpha + w_i\tau + x_{i1}\beta_0 + w_i(x_{i1} - E[x_1])\beta_1 \\ &+ f_{v_0}(z_i)h(w_i, z_i) + w_i f_{v_1}(z_i)h(w_i, z_i) + \eta_i \end{aligned} \quad (2.10)$$

where $E[\eta_i|z_i, w_i] = 0$ by construction.

Note that in our estimating Equation (2.10) the covariates appear by themselves, interacted with the generalized errors, and interacted with both w_i and the generalized error. The excluded instruments do not appear on their own but only interacted with w_i and the generalized error. In addition, the generalized errors are interacted with both the covariates in the outcome equation x_{i1} as well as the excluded instruments x_{i2} .

The first four terms in our estimating equation would appear in standard regression adjustment analysis when the treatment, w_i could be assumed to be unconfounded conditional on x_{i1} ; and instrumental variables would not be needed. The last two terms account for the potential endogeneity of w_i by assuming z_i has at least one element not in x_{i1} , and that element explains sufficient variation in the treatment. These two terms constitute correction functions that are flexible functions of the vector of exogenous variables, z_i . Finally, the generalized error $h(w_i, z_i)$ comes from a heteroskedastic probit model. It is important to note here that the functions $f_{v_0}(z_i)$ and $f_{v_1}(z_i)$ have no restrictions on them and one can specify them as flexible functions of z_i .⁵

3 | Estimation

For estimation, we have two equations given by (2.3) and (2.10):

$$\begin{aligned} w_i &= \mathbb{1}[z_i\gamma + v_i \geq 0] \\ y_i &= \alpha + w_i\tau + x_{i1}\beta_0 + w_i(x_{i1} - E[x_1])\beta_1 \\ &+ f_{v_0}(z_i)h(w_i, z_i) + w_i f_{v_1}(z_i)h(w_i, z_i) + \eta_i. \end{aligned}$$

These two equations suggest a two-step estimation procedure. In the first step, we estimate the reduced form model of the treatment variable or the switching indicator to obtain the generalized residuals, $\hat{h}(w_i, z_i)$; these are estimators of the generalized errors, $h(w_i, z_i)$. In the second step, we plug-in the estimates from the first stage and we have

$$\begin{aligned} y_i &= \alpha + \tau w_i + x_{i1}\beta_0 + w_i(x_{i1} - \bar{x}_1)\beta_1 + f_{v_0}(z_i)\hat{h}(w_i, z_i) \\ &+ w_i f_{v_1}(z_i)\hat{h}(w_i, z_i) + e_i. \end{aligned}$$

Here, the error term e_i includes η_i but also the estimation error in $\hat{h}(w_i, z_i)$ and the sampling error in $\bar{x}_1$. In the second step, we estimate τ along with the parameters $\alpha_0, \beta_0, \beta_1$. In the second step, we also estimate the functions $f_{v_0}(z_i)$ and $f_{v_1}(z_i)$ and they may be parametric or nonparametric. If we specify the functions $\{\sigma_v(z_i), f_{v_0}(z_i), f_{v_1}(z_i)\}$ to be parametric, the estimation procedure is an extension of the usual two-step Heckman method that allows for heteroskedasticity in the treatment equation and also, just as importantly, heterogeneity in the coefficients of the conditional linear projection. We do not impose further distributional assumptions on the unobservables.

We now describe parametric modeling of $\{\sigma_v(z_i), f_{v_0}(z_i), f_{v_1}(z_i)\}$ and then cover the case where these functions are unspecified.

3.1 | Parametric Estimation

For the parametric estimation of the model parameters, we first specify the functional forms of unknown functions of our model: $\sigma_v(z_i) = \sigma_v(z_i, \Pi)$ and for $f_{v_0}(z_i) = f_{v_0}(z_i, \Gamma_0)$ and $f_{v_1}(z_i) = f_{v_1}(z_i, \Gamma_1)$. A popular heteroskedasticity function is $\sigma_v^2(z_i) \equiv \exp(2S(z_i)\Pi)$, where $S(z_i) \equiv (s_1(z_i), \dots, s_{K_3}(z_i))$ is a $1 \times K_3$ vector of covariates, excluding an intercept, and Π is the corresponding parameter vector of dimension $K_3 \times 1$. We also specify $f_{v_j}(z_i) \equiv F_{v_j}(z_i)\Gamma_j, j = \{0, 1\}$ for all $i = 1, \dots, N$.⁶ $\{F_{v_0}(\cdot) \equiv (f_{01}(z_i), \dots, f_{0K_4}(z_i))\}$ is a $1 \times K_4$ vector of covariates and Γ_0 is the corresponding parameter vector of dimension $K_4 \times 1$. Similarly, $\{F_{v_1}(\cdot) \equiv (f_{11}(z_i), \dots, f_{1K_5}(z_i))\}$ is a $1 \times K_5$ vector of covariates and Γ_1 is the corresponding parameter vector of dimension $K_5 \times 1$. We can include flexible functions including interactions, squares, logarithms and other functional forms of the explanatory variables in $F_{v_0}(\cdot), F_{v_1}(\cdot)$ and $S(\cdot)$.

Given these parametric specifications, we can describe a two-step parametric estimation procedure. The true parameters to be estimated in the first stage can be denoted by $\theta_{w*} \equiv \{\gamma_*, \Pi_*\}$ that belong to a finite dimensional parameter space Θ_w . In the second step, we denote the full set of true parameters to be estimated as $\theta_{y*} \equiv \{\alpha_*, \tau_*, \beta_{0*}, \beta_{1*}, \Gamma_{0*}, \Gamma_{1*}\}$ that belong to the finite dimensional parameter space Θ_y . Procedure 3.1 summarizes the estimation method.

Procedure 3.1. (Parametric Estimation). Under Assumptions 2.1–2.4, the parameters are consistently estimated in the following two-step estimation method:

1. Specify a parametric functional form for $\sigma_v^2(z_i)$ as $\sigma_v^2(z_i, \Pi)$. Using all N observations, run a heteroskedastic probit⁷ by maximizing the log likelihood

$$\sum_{i=1}^N \left[w_i \log \left[\Phi \left(\frac{z_i \gamma}{\sigma_v(z_i, \Pi)} \right) \right] + (1 - w_i) \log \left[1 - \Phi \left(\frac{z_i \gamma}{\sigma_v(z_i, \Pi)} \right) \right] \right] \quad (3.1)$$

to obtain $\hat{\theta}_w \equiv (\hat{\gamma}, \hat{\Pi})$, where $\Phi(\cdot)$ denotes standard normal cdf. Obtain the generalized residuals

$$\hat{h}_i \equiv \left[w_i \lambda \left(\frac{z_i \hat{\gamma}}{\hat{\sigma}_v(z_i)} \right) - (1 - w_i) \lambda \left(-\frac{z_i \hat{\gamma}}{\hat{\sigma}_v(z_i)} \right) \right]. \quad (3.2)$$

2. Specify a functional form for $f_{v_0}(z_i) = f_{v_0}(z_i, \Gamma_0)$ and $f_{v_1}(z_i) = f_{v_1}(z_i, \Gamma_1)$. Using all N observations, estimate the equation

$$y_i = \alpha + w_i \tau + x_{i1} \beta_0 + w_i (x_{i1} - \bar{x}_1) \beta_1 + \hat{h}_i f_{v_0}(z_i, \Gamma_0) + w_i \hat{h}_i f_{v_1}(z_i, \Gamma_1) + e_i \quad (3.3)$$

using OLS, where $\bar{x}_1$ denotes the sample mean. $\square$

Supporting Information: Appendix A describes the asymptotic properties of the parametric estimator. Large sample properties of the parametric estimators are obtained by standard application of the generalized method of moments arguments applied to two-step estimation problems see, for example, Wooldridge (2010), section 14.1.⁸

One benefit of using CLPs to approximate the conditional expectations is that it allows us to describe a semiparametric estimation procedure. We describe a semiparametric estimation procedure where we use the method of sieves in both steps. Formally introduced by Grenander (1981), the method of sieves allows for unknown functions that lie in an infinite-dimensional parameter space to be approximated by finite-dimensional spaces of growing dimensions. Consistency is achieved by allowing the dimensionality to grow with sample size. Sieve estimation is easily implementable as for a given sample size, it is essentially a flexible parametric estimation: in standard softwares, sieve estimation is implemented by manually adding sieve terms as additional covariates. For inference, similar to the parametric case, an important consideration in a two-step estimation procedure is whether and how the estimation error of the first step estimators affects the asymptotic variance of the second step estimators. In the parametric case, the methods of adjusting the asymptotic variance of the second step estimator follows the standard procedures given in Wooldridge (2010). In the sieve case, recent work in the asymptotics of two-step sieve estimation by Hahn et al. (2018) makes the large sample inference of our estimator possible: our two-step sieve estimation procedure fits neatly into their framework and thus the asymptotic results are directly applied to obtain the asymptotic normality result. Additionally, they also derive the numerical equivalence result that facilitates

inference in practice: we can treat the two-step sieve estimation as if it were a standard two-step parametric estimation and thus conduct a practical inference on the parameters.

3.2 | Two-Step Sieve Estimation

To describe our two-step sieve plug-in estimation procedure, denote the data in the first step by Z_{wi} and data in the second step by Z_{yi} . Let Z_i be the union of the distinct elements of Z_{wi} and Z_{yi} and assume $\mathfrak{F}_z$ to be the distribution of Z_i . $\{Z_i\}_{i=1}^N$ is assumed to be a sequence of strictly stationary weakly dependent data. Let $(\Theta_w, \|\cdot\|_{\Theta_w})$ and $(\Theta_y, \|\cdot\|_{\Theta_y})$ be some infinite dimensional separable complete metric spaces and $\|\cdot\|_{\Theta_w}, \|\cdot\|_{\Theta_y}$ could be sup-norm, the $L_2(d\mathfrak{F}_z)$ norm, the Sobolev norm or the Holder norm. Let $\theta_{w*} \equiv \{\gamma_*, \sigma_{v*}^2(Z_i)\} \in \Theta_w$ denote the true parameters in the first step and let $Q_{wi} \equiv Q_w(\theta_w; Z_{wi}) \equiv w_i \log \left[\Phi \left(\frac{Z_i \gamma}{\sigma_v(Z_i)} \right) \right] + (1 - w_i) \log \left[1 - \Phi \left(\frac{Z_i \gamma}{\sigma_v(Z_i)} \right) \right]$. Let $\beta_* \equiv (\theta_*, \alpha_*, \tau_*, \beta'_{0*}, \beta'_{1*})'$, $f_{v*}(Z_i) \equiv (f_{v_0*}(Z_i), f_{v_1*}(Z_i))$. Further define $X_i \equiv (x_{i1}, w_i(x_i - E[x_1]))$, $H_i \equiv H(\theta_{w*}; Z_i) \equiv [h(\theta_{w*}; Z_i), w_i h(\theta_{w*}; Z_i)]'$. In the second stage, let $\theta_{y*} \equiv (\beta'_*, f_{v*}(Z_i)) \in \Theta_y$ denote the true parameters and define $Q_{yi} \equiv Q_y(\theta_y, \theta_{w*}; z_{yi}) \equiv -\frac{[y_i - (X_i \beta + f_{v*}(Z_i) H_{*i})]^2}{2}$.

In our two-step sieve M estimation, we restrict the estimators of θ_{w*} and θ_{y*} to be in some finite dimensional sieve spaces denoted by Θ_{wN} and Θ_{yN} respectively. The preliminary step in both steps is to define the basis functions and the sieve spaces for the unknown functions in both the steps. Specifically, in the first step, we define a sieve space for $\sigma_{v*}^2(Z_i)$ and in the second step, we define the sieve spaces for $\{f_{v_0*}(Z_i), f_{v_1*}(Z_i)\}$. We assume that the collection of all the unknown functions in our model $[\sigma_{v*}^2(Z_i), f_{v_0*}(Z_i), f_{v_1*}(Z_i)]$ belongs to a Hölder class. This allows the unknown functions to be easily estimated by linear sieves. The following procedure describes our two-step sieve M estimation.

Procedure 3.2. (Sieve Estimation)

1. In the first stage, our true parameters to be estimated are $\{\gamma_*, \sigma_{v*}^2(Z_i)\}$ that lie in the infinite-dimensional parameter space Θ_w . Define the finite-dimensional sieve space as

$$\Theta_{wN} = \mathbb{B}_w \times \left\{ \sigma_{vN}(\cdot) = \exp(S_{K_{3,N}}(Z_i) \Pi_N) : \Pi_N \in \mathbb{R}^{K_{3,N}} \right\} \quad (3.4)$$

where $\times$ denotes the Cartesian product. Denote the dimension of Θ_{wN} by $K_w(N)$. $\mathbb{B}_w$ is the finite dimensional parameter space for γ_* and $S_{K_{3,N}}(Z_i) \equiv (s_1(Z_i), \dots, s_{K_{3,N}}(Z_i))$ is a $1 \times K_{3,N}$ vector of basis functions. Next, denoting $\Phi(\cdot)$ as standard normal cdf, our first step sieve estimator $\hat{\theta}_{wN}$ is defined as

$$\begin{aligned} & \frac{1}{N} \sum_{i=1}^N Q_{wi}(\hat{\theta}_{wN}; Z_{wi}) \\ & \geq \sup_{\theta_{wN} \in \Theta_{wN}} \frac{1}{N} \sum_{i=1}^N Q_{wi}(\theta_{wN}; Z_{wi}) - O_p(\epsilon_{1,N}^2) \end{aligned} \quad (3.5)$$

where $\epsilon_{1,N}^2$ is the optimization error that goes to zero as $N \rightarrow \infty$.

In practice, this amounts to running a flexible heteroskedastic probit that maximizes the following log-likelihood:

$$\begin{aligned} \hat{\theta}_{wN} = \arg \max_{\theta_{wN} \in \Theta_{wN}} \sum_{i=1}^N \left[w_i \log \left[\Phi \left(\exp(-S_{K_{3,N}}(Z_i) \Pi_N) Z_{wi} \gamma \right) \right] \right. \\ \left. + (1 - w_i) \log \left[1 - \Phi \left(\exp(-S_{K_{3,N}}(Z_i) \Pi_N) Z_{wi} \gamma \right) \right] \right] \end{aligned} \quad (3.6)$$

using all N observations. This would give us estimators for γ_* and $\sigma_v^2(Z_i)$ which would subsequently give us an estimator of $\lambda(\cdot)$ and $h(Z_i, w_i)$. Denote the first stage estimators as $\hat{\gamma}$, $\hat{\sigma}_v^2(Z_i)$, $\hat{\lambda}_i \equiv \lambda\left(\frac{Z_i \hat{\gamma}}{\hat{\sigma}_v(Z_i)}\right)$ and $\hat{h}_i$.

2. In the second stage, we have true parameters $\{\alpha_*, \tau_*, \beta_{0*}, \beta_{1*}, f_{v_0^*}(Z_i), f_{v_1^*}(Z_i)\}$ that lie in the infinite parameter space Θ_y . For sieve estimation in the second step, define the finite-dimensional sieve space as

$$\Theta_{yN} \equiv \mathbb{B}_y \times \left\{ f_{v_0N}(\cdot) = F_{0K_{4,N}}(Z_i) \Gamma_{0,N} : \Gamma_{0,N} \in \mathbb{R}^{K_{4,N}} \right\} \quad (3.7)$$

$$\begin{aligned} \times \left\{ f_{v_1N}(\cdot) = F_{1K_{5,N}}(Z_i) \Gamma_{1,N} : \Gamma_{1,N} \in \mathbb{R}^{K_{5,N}} \right\} \quad (3.8) \\ = \mathbb{B}_y \times \mathbb{F}_0 \times \mathbb{F}_1. \end{aligned}$$

Denote the dimension of Θ_{yN} by $K_y(N)$. $\mathbb{B}_y$ are the finite-dimensional parameter spaces for the finite true parameters $(\alpha_*, \tau_*, \beta_{0*}, \beta_{1*})$. $F_{0K_{4,N}}(Z_i) \equiv \{f_{0,1}(\cdot), \dots, f_{0,K_{4,N}}(\cdot)\}$ is $1 \times K_{4,N}$ vector of basis functions and $F_{1K_{5,N}}(Z_i) \equiv \{f_{1,1}(\cdot), \dots, f_{1,K_{5,N}}(\cdot)\}$ is $1 \times K_{5,N}$ vector of basis functions. Denoting $\{\hat{f}_{v_0}(Z_i), \hat{f}_{v_1}(Z_i)\} \equiv \{F_{0K_{4,N}}(Z_i) \hat{\Gamma}_{0,N}, F_{1K_{5,N}}(Z_i) \hat{\Gamma}_{1,N}\}$, sieve estimator in the second step $\hat{\theta}_{yN} \equiv \{\hat{\alpha}, \hat{\tau}, \hat{\beta}_0, \hat{\beta}_1, \hat{f}_{v_0}(Z_i), \hat{f}_{v_1}(Z_i)\} \in \Theta_{yN}$ is defined as

$$\begin{aligned} \frac{1}{N} \sum_{i=1}^N Q_y(\hat{\theta}_{yN}, \hat{\theta}_{wN}; Z_{yi}) \\ \geq \sup_{\theta_{yN} \in \Theta_{yN}} \frac{1}{N} \sum_{i=1}^N Q_y(\theta_{yN}, \hat{\theta}_{wN}; Z_{yi}) - O(\epsilon_{2,N}^2) \end{aligned} \quad (3.9)$$

where $\epsilon_{2,N}^2$ is the optimization error that goes to zero as $N \rightarrow \infty$.

Thus the second step sieve estimator $\hat{\theta}_{yN}$ is obtained via the least square of

$$\begin{aligned} y_i = \alpha + w_i \tau + x_{i1} \beta_0 + w_i (x_{i1} - \bar{x}_1) \beta_1 + \hat{h}_i F_{0K_{4,N}}(Z_i) \Gamma_{0,N} \\ + w_i \hat{h}_i F_{1K_{5,N}}(Z_i) \Gamma_{1,N} + e_i. \end{aligned} \quad (3.10)$$

This amounts to running a flexible regression of y_i on $1, x_{i1}, w_i(x_{i1} - \bar{x}_1), \hat{h}_i F_{0K_{4,N}}(Z_i), \hat{h}_i w_i F_{1K_{5,N}}(Z_i)$ using all N observations. □

Supporting Information: Appendix B describes the asymptotic properties of the sieve estimator where we use the general unified mathematical framework for asymptotic properties of sieve two-step estimators described in Hahn et al. (2018). Since our

sieve estimation procedure neatly fits into their framework, our estimator directly follows their results. They also show the numerical equivalence result between parametric variances and sieve variances that we use to obtain expressions for variances of our two-step sieve estimator.

3.3 | A Note on Identification

Our two-step estimation procedures require us to specify functions for $\sigma_v^2(z_i)$ in the first step and functions for $f_{v_0}(z_i)$ and $f_{v_1}(z_i)$ in the second step. Correspondingly, while the functional forms for these functions are mostly unrestricted, it would be useful to discuss some themes that would be key for identification.

For the parametric estimation procedure, identification in the first step depends on the standard full rank assumptions on the covariate matrix and additionally on the functional form specified for $\sigma_v^2(z_i)$. Since variance is nonnegative, the specification for $\sigma_v^2(z_i)$ should also be a nonnegative function. An exponential function with a linear index without an intercept is a popular choice. However, one needs to be a little careful as Carlson (2019) shows that certain parametric heteroskedastic functions can result in a lack of identification in a multiplicative heteroskedastic model. In the second step, we specify functional forms for two unknown functions $f_{v_0}(z_i)$ and $f_{v_1}(z_i)$ and though both linear and nonlinear specifications are allowed in the CLPs, identification might fail for some rare nonlinear specifications and there are some things that we need to be mindful of.

To see this, suppose we have $x_i \equiv \{1, x_{i1}\}$ and $z_i \equiv \{1, x_{i1}, x_{i2}\}$ for scalar x_{i1} and x_{i2} . Let $\dot{x}_{i1} \equiv x_{i1} - \bar{x}_1$ where $\bar{x}_1$ is the sample mean and $h_i \equiv h(w_i, z_i)$. Note that in the simplest case, we can specify $f_{v_0}(z_i), f_{v_1}(z_i)$ as linear functions where $f_{v_0}(z_i) = \rho_{00} + x_{i1} \rho_{01} + x_{i2} \rho_{02}$ and $f_{v_1}(z_i) = \rho_{10} + x_{i1} \rho_{11} + x_{i2} \rho_{12}$. We get

$$\begin{aligned} y_i = \alpha + \tau w_i + x_{i1} \beta_0 + \dot{x}_{i1} w_i \beta_1 + \rho_{00} h_i + x_{i1} h_i \rho_{01} + x_{i2} h_i \rho_{02} \\ + w_i h_i \rho_{10} + x_{i1} w_i h_i \rho_{11} + x_{i2} w_i h_i \rho_{12} + e_i \end{aligned} \quad (3.11)$$

that is identified. Certain specifications for $f_{v_0}(z_i), f_{v_1}(z_i)$, that are nonlinear in parameters; for example, $f_{v_0}(z_i)$ specified as $(x_{i1} - \rho_0)^2$; would lead to

$$y_i = \alpha + \tau w_i + x_{i1} \beta_0 + \dot{x}_{i1} w_i \beta_1 + (x_{i1} - \rho_0)^2 h_i + w_i h_i f_{v_1}(z_i) + \eta_i \quad (3.12)$$

where we have nonidentification⁹ due to the possibility of multiple solutions (Lewbel 2019).

Identification will also fail if the conditional expectations in Assumption 2.4 include z_i as a linear function, for example, as $E[u_{ij}|v_i, z_i] = \psi_{j0} + \psi_{jv} v_i + x_{i1} \psi_{j1} + x_{i2} \psi_{j2}$. In this case, we will get

$$\begin{aligned} E[y_i|z_i, v_i] = \alpha + w_i \tau + x_{i1} \beta_0 + w_i (x_{i1} - E[x_1]) \beta_1 \\ + [\psi_{00} + \psi_{0v} v_i + x_{i1} \psi_{01} + x_{i2} \psi_{02}] \\ + w_i [\psi_{10} + \psi_{1v} v_i + x_{i1} \psi_{11} + x_{i2} \psi_{12}] \end{aligned}$$

and then

$$E[y_i|z_i, w_i] = \alpha + w_i\tau + x_{i1}\beta_0 + w_i(x_{i1} - E[x_1])\beta_1 \\ + \psi_{00} + \psi_{0v}h_i + x_{i1}\psi_{01} + x_{i2}\psi_{02} + w_i\psi_{10} \\ + \psi_{1v}w_ih_i + w_ix_{i1}\psi_{11} + w_ix_{i2}\psi_{12}.$$

Even if we exclude a constant in the specifications, we still get

$$E[y_i|z_i, v_i] = \alpha + w_i\tau + x_{i1}\beta_0 + w_i(x_{i1} - E[x_1])\beta_1 \\ + [\psi_{0v}v_i + x_{i1}\psi_{01} + x_{i2}\psi_{02}] \\ + w_i[\psi_{1v}v_i + x_{i1}\psi_{11} + x_{i2}\psi_{12}]$$

and then

$$E[y_i|z_i, w_i] = \alpha + w_i\tau + x_{i1}\beta_0 + w_i(x_{i1} - E[x_1])\beta_1 \\ + \psi_{0v}h_i + x_{i1}\psi_{01} + x_{i2}\psi_{02} + \psi_{1v}w_ih_i \\ + w_ix_{i1}\psi_{11} + w_ix_{i2}\psi_{12}$$

where the parameters are no longer identified.

The two-step sieve estimation procedure is based in the identifying assumption that $\theta_{w*} \in \Theta_w$ is the unique solution to $\sup_{\theta_w \in \Theta_w} E[Q_w(\theta_w; Z_{wi})]$ and $\theta_{y*} \in \Theta_y$ is the unique solution to $\sup_{\theta_y \in \Theta_y} E[H_w(\theta_w, \theta_y; Z_{wi})]$ where $(\Theta_w, \|\cdot\|_{\Theta_w})$ and $(\Theta_y, \|\cdot\|_{\Theta_y})$ are some infinite dimensional separable complete metric spaces, and $\|\cdot\|_{\Theta_w}$ and $\|\cdot\|_{\Theta_y}$ could be sup-norm, the $L_2(d\mathcal{F}_Z)$ norm, the Sobolev norm or the Hölder norm. With respect to nonparametrically specifying the unknown functions $\sigma_{v*}^2(z_i)$, $f_{v*0}(z_i)$, $f_{v*1}(z_i)$, adhering to the similar considerations as in the parametric estimation procedure discussed above would help to keep the sieve estimation tractable. For example, while we are no longer restricted to enclose the basis functions within the exponential function as long as we use nonnegative sieves,¹⁰ using the exponential function would keep estimation easily applicable by using standard statistical packages for heteroskedastic probit (such as `hetprobit` command in Stata). In the second step, we have even more flexibility as we can use polynomials, splines, B-splines to specify $f_{v*0}(z_i)$, $f_{v*1}(v_i)$, though it would be useful to avoid certain unconventional, fringe specifications to avoid nonidentification.

TABLE 1 | Simulation results.

	DGP 1			DGP 2		
	Bias	SD	RMSE	Bias	SD	RMSE
OLS	0.43843	0.01217	0.4386	0.88568	0.01078	0.88575
IV	0.00081	0.02507	0.02508	0.08838	0.03936	0.09674
ESM	0.00057	0.01742	0.01742	0.33108	0.03576	0.33301
Procedure 3.1	0.00062	0.02144	0.02145	0.00109	0.02464	0.02466
Procedure 3.2: Polynomial sieve	0.00092	0.02589	0.0259	0.00207	0.03815	0.03819
Procedure 3.2: Spline sieve	0.00096	0.02611	0.02612	0.0026	0.0383	0.03838

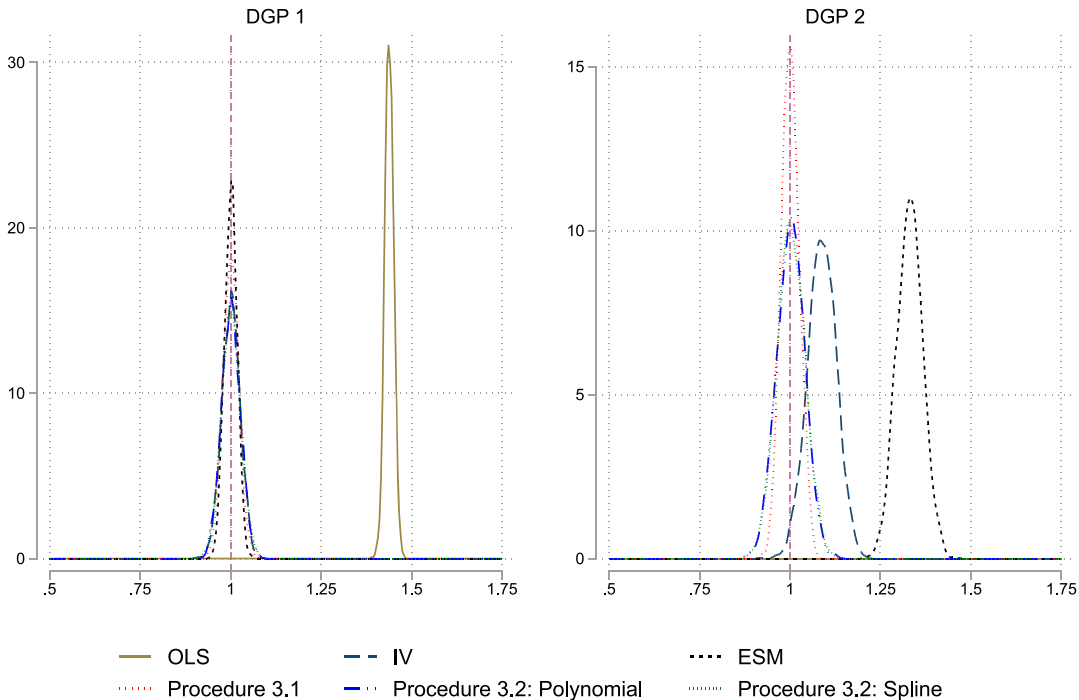


FIGURE 1 | Kernel smoothed empirical distributions of estimators.

4 | Monte Carlo Simulations

In this section, we provide Monte Carlo simulations to illustrate the finite sample properties of our proposed estimator. We find that when heteroskedasticity is present in the treatment assignment model, only our proposed estimator has the smallest bias and least Root Mean Squared Error (RMSE) compared with standard estimation procedures. The exogenous variables x_{i1} , x_{i2} are generated as $x_{i1} \sim \text{Normal}(0,1)$ and $x_{i2} \sim \text{Normal}(0.7x_{i1}, 0.5)$. The potential outcomes are generated as in equation

$$y_i^0 = \alpha^0 + x_{i1}\beta^0 + u_i^0 \quad (4.1)$$

$$y_i^1 = \alpha^1 + x_{i1}\beta^1 + u_i^1 \quad (4.2)$$

where $\alpha^0 = 1$, $\alpha^1 = 2$, $\beta^0 = 2$, $\beta^1 = 3$ and the switching variable is generated as

$$w_i = \mathbb{1}[\gamma_0 + x_{i1}\gamma_1 + x_{i2}\gamma_2 + v_i \geq 0] \quad (4.3)$$

where $\gamma_0 = 0$, $\gamma_1 = 1$, $\gamma_2 = 2$. The unobservables u_i^0, u_i^1, v_i in the model vary across the designs. Let e_{i0}, e_{i1}, e_{i3} be drawn from the following multivariate normal distribution:

$$\begin{pmatrix} e_{i1} \\ e_{i2} \\ e_{i3} \end{pmatrix} \sim \text{Normal} \left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0.5 & 0.2 \\ 0 & 1 & 0.75 \\ 0 & 0 & 1 \end{bmatrix} \right) \quad (4.4)$$

1. DGP 1: In this DGP, the error in the treatment assignment model is homoskedastic. So we have $u_i^0 = e_{i1}$, $u_i^1 = e_{i2}$ and $v_i = e_{i3}$.
2. DGP 2: In this DGP, the error in the treatment assignment model is heteroskedastic. We have $u_i^0 = e_{i1}$, $u_i^1 = e_{i2}$ and $v_i = \exp(1.5x_{i2})e_{i3}$.

TABLE 2 | Descriptive statistics, NHANES 2007–2016.

Variables	Mean/proportion		
	Overall	Was advised to lose weight	Was not advised to lose weight
Calorie intake (kcal)	2696.701	2639.031	2735.629
Advice (%)	40.32		
Age (years)	48.02	50.43718	46.39
Female	36.48	43.01	32.06
Hispanic (%)	15.05	13.83	15.87
White (%)	68.76	68.28	69.09
Black (%)	10.27	11.80	9.24
Other (%)	5.92	6.09	5.81
Less than 9 years of schooling (%)	4.18	3.55	4.60
9th to 12 years of schooling (%)	9.24	8.33	9.86
High School grad or some equivalent (%)	23.18	22.50	23.64
Some college or AA (%)	33.10	34.61	32.07
College and above (%)	30.30	31.01	29.83
Married (%)	60.02	62.38	58.43
BMI	31.95	34.75	30.06
Is employed (%)	67.67	64.95	69.51
Monthly income less than \$2100 (%)	25.25	22.65	27.01
Monthly income between \$2100 and \$5400 (%)	30.72	30.10	31.14
Monthly income between \$5400 and \$8400 (%)	17.92	20.11	16.44
Monthly income more than \$8400 (%)	18.53	18.07	18.84
High BP (%)	39.25	54.96	28.63
Cholesterol (%)	48.66	66.11	36.87
Heart condition (%)	3.51	5.07	2.46
Diabetes (%)	13.55	23.89	6.57
Arthritis (%)	27.83	37.84	21.07
Health insurance (%)	83.34	89.29	79.33
Routine care (%)	85.45	93.27	80.16
Number of observations	6350	2698	3652

Note: Survey cycles: 2007–2008, 2009–2010, 2013–2014, and 2015–2016. The data are pooled data from all the survey cycles. Results are weighted to be nationally representative.

Data Source: National Health and Nutrition Survey (NHANES).

TABLE 3 | Average partial effects.

Advice	(1) Probit	(2) Het Probit Parametric	(3) Het Probit Sieve
Age	−0.002** (0.001)	−0.002*** (0.001)	−0.002*** (0.001)
Female	0.077*** (0.019)	0.083*** (0.019)	0.084*** (0.019)
Hispanic	0.091*** (0.029)	0.083*** (0.028)	0.083*** (0.028)
Black	0.052* (0.026)	0.040 (0.027)	0.040 (0.027)
Other	0.078** (0.038)	0.081** (0.038)	0.081** (0.038)
9th grade	−0.021 (0.044)	−0.034 (0.044)	−0.035 (0.044)
High School / GED	0.018 (0.042)	0.008 (0.041)	0.007 (0.041)
Some College / AA	0.042 (0.040)	0.028 (0.040)	0.027 (0.040)
College and above	0.090** (0.043)	0.076* (0.042)	0.075* (0.042)
Married	−0.009 (0.020)	−0.008 (0.020)	−0.009 (0.020)
BMI	0.035*** (0.002)	0.045*** (0.002)	0.046*** (0.003)
Employed	0.030 (0.022)	0.032 (0.023)	0.032 (0.023)
Income < \$2100	−0.038 (0.037)	−0.030 (0.036)	−0.031 (0.036)
\$2100 ≤ Income ≤ \$5400	−0.005 (0.035)	−0.014 (0.034)	−0.014 (0.034)
\$5400 ≤ Income ≤ \$8400	0.031 (0.035)	0.020 (0.036)	0.020 (0.036)
High BP	0.127*** (0.024)	0.126*** (0.024)	0.125*** (0.024)
Cholesterol	0.200*** (0.025)	0.213*** (0.025)	0.214*** (0.025)
Heart Condition	0.040 (0.043)	0.026 (0.042)	0.026 (0.042)
Diabetes	0.223*** (0.035)	0.227*** (0.033)	0.228*** (0.033)
Arthritis	0.072*** (0.024)	0.064*** (0.023)	0.064*** (0.023)
Health Insurance	0.094*** (0.026)	0.103*** (0.027)	0.104*** (0.027)

(Continues)

TABLE 3 | (Continued)

Advice	(1) Probit	(2) Het Probit Parametric	(3) Het Probit Sieve
Routine Care	0.166*** (0.032)	0.175*** (0.034)	0.175*** (0.034)
Observations	6,350	6,350	6,350
Survey Cycle Fixed Effects	YES	YES	YES
F-stat (instruments)	23.79***	22.53***	22.28***

Note: Montiel–Pflueger effect F-stat weak IV test = 51.708***. LR test statistic for heteroskedasticity = 74.84. Wald test statistic for heteroskedasticity = 67.95***

Data source: Pooled data from the National Health and Nutrition Survey (NHANES) survey cycles 2007–2008, 2009–2010, 2013–2014, and 2015–2016. Results weighted to be nationally represented. Estimates are obtained from probit, heteroskedastic probit regressions weighted to account for the complex survey design. All models include indicators for survey cycles. Stars (*0.10, **0.05, and ***0.01).

We compare the finite sample behavior of our proposed estimator in Procedure 3.1 and Procedure 3.2 with OLS, IV, ESM estimators. We consider two kinds of sieves for estimation in Procedure 3.2 polynomial sieve and spline sieve. Simulations are run for $N = 5000$, and results are gathered for 2000 simulations.

4.1 | Results

Table 1 reports the bias, standard deviation, and RMSE for the estimates of τ . Figure 1 displays the kernel smoothed empirical distributions of the estimates for τ from the 1000 simulations. As we can see, not surprisingly, without heteroskedasticity present, the traditional ESM estimator has the smallest RMSE. When heteroskedasticity is present, only our proposed estimator has the smallest bias and RMSE compared with all other considered estimators and performs considerably better.

5 | Empirical Application: The Effect of Physician Advice on Calorie Consumption

According to National Health and Nutrition Examination Survey (NHANES), the prevalence of obesity and severe obesity in the United States has increased significantly among adult men and women between 1999–2000 and 2017–2018 and 42.4% of US adults had obesity in 2017–2018 (NIDDK 2021). The US Centers of Disease Control and Prevention (CDC) estimates the annual obesity-related medical care costs in the United States to be nearly \$173 billion in 2019 dollars (CDC 2024)

In light of this, the US Preventive Services Task Force (USPSTF) recommends that all patients be screened for obesity and, if needed, be provided weight-loss advice (USPSTF 2018). Research has supported physician advice to be a lower-cost and more precisely targeted public health interventions helpful in influencing consumer decisions such as smoking, diet, drinking, and exercise (Ippolito and Mathios 1991, 1995; Kenkel 1990; Kenkel and Terza 2001). Kenkel and Terza (2001) found physician advice to have substantial and significant impact on alcohol consumption.

Research investigating the impact of physician advice as public health tool for controlling obesity is limited. Research has shown

that physician advice to lose weight has positive effects on the efforts to lose weight (Jackson et al. 2013; Loureiro and Nayga Jr, 2006). For example, Loureiro and Nayga (2006) found that physician advice to lose weight has positive effects on both the probability of eating fewer calories and fat and using exercise to lose weight. However, they do not account for the endogeneity of receiving such weight-loss advice from physician. Moreover, Lorts and Ohri-Vachaspati (2016) describe the determinants of receiving weight-loss advice and find substantial disparities in who receives weight-loss advice. For example, they find that individuals in the lowest income group had significantly lower odds than individuals in higher income groups of receiving weight-loss advice from their health care provider. White and Stubblefield-Tave (2017) discuss that while such disparities in health care are partly due to socioeconomic factors such as income, nonsocioeconomic factors such as unconscious bias, stereotyping, racism, gender bias also play a large contributory role. To the best of our knowledge, previous research evaluating the impact of physician advice on health outcome has not accounted for these factors. Our paper contributes to this discourse in two key ways. First, we allow whether or not an individual receives advice to lose weight to be endogenous. Second, we allow for the heterogeneity in the propensity to receive the advice through incorporating heteroskedastic treatment assignment model in the ESM framework.

5.1 | Data

We estimate the effect of physician advice on calorie consumption using the data from the five most recent cycles of the National Health and Nutritional Examination Survey (NHANES). NHANES is a cross-sectional national survey that represents a stratified multistage probability sample of the noninstitutionalized US population. NHANES comprises of questionnaires, physical examinations and laboratory samples. For our analysis, we combine the data from the 2007–2008, 2009–2010, 2011–2012, 2013–2014, and 2015–2016 surveys to form a pooled cross section. We exclude pregnant women and observations with missing data. We specifically look at overweight individuals and thus limit our sample to those individuals who have BMI equal to or more than 25.0 between the age of 20 and 80 years. We also exclude the individuals who report consuming less than 1800

TABLE 4 | Comparison of estimators of calorie consumption.

Log(Calories)	(1)	(2)	(3)	(4)	(5)	(6)
	OLS	2SLS	ESM	Proc 3.1	Proc 3.2 Poly	Proc 3.2 Spline
Advice	0.018 (0.016)	0.567 (0.903)	−0.092* (0.054)	−0.143** (0.062)	−0.144** (0.074)	−0.146** (0.071)
Female	−0.151*** (0.011)	−0.131** (0.063)	−0.148*** (0.012)	−0.152*** (0.013)	−0.151*** (0.013)	−0.149*** (0.013)
Hispanic	0.000 (0.012)	−0.022 (0.043)	−0.014 (0.014)	−0.015 (0.015)	−0.014 (0.015)	−0.011 (0.015)
Black	−0.001 (0.013)	0.026 (0.055)	−0.008 (0.015)	−0.009 (0.015)	−0.008 (0.015)	−0.006 (0.015)
Other	−0.053*** (0.018)	−0.074 (0.078)	−0.061*** (0.019)	−0.063*** (0.019)	−0.062*** (0.019)	−0.060*** (0.019)
9th grade	0.060** (0.026)	0.090 (0.073)	0.049* (0.027)	0.051* (0.027)	0.051* (0.027)	0.052* (0.027)
High School / GED	0.037* (0.020)	0.010 (0.055)	0.027 (0.024)	0.027 (0.024)	0.027 (0.024)	0.029 (0.024)
Some College / AA	0.020 (0.019)	0.040 (0.063)	0.007 (0.024)	0.007 (0.024)	0.007 (0.024)	0.008 (0.024)
College and above	0.023 (0.019)	−0.085 (0.089)	0.018 (0.026)	0.014 (0.026)	0.015 (0.026)	0.021 (0.026)
Married	−0.033*** (0.011)	0.036 (0.043)	−0.019 (0.013)	−0.019 (0.013)	−0.019 (0.013)	−0.019 (0.013)
High BP	−0.017 (0.014)	0.054 (0.104)	0.013 (0.015)	0.006 (0.017)	0.008 (0.017)	0.013 (0.017)
Cholesterol	−0.034** (0.015)	−0.061 (0.080)	−0.000 (0.016)	−0.009 (0.019)	−0.006 (0.020)	0.002 (0.020)
Arthritis	−0.011 (0.014)	0.064 (0.147)	0.024 (0.015)	0.021 (0.016)	0.022 (0.016)	0.026 (0.016)
Heart Condition	−0.045 (0.032)	0.205 (0.203)	−0.011 (0.032)	−0.010 (0.033)	−0.009 (0.033)	−0.007 (0.033)
Diabetes	0.011 (0.016)	0.206 (0.152)	0.040 (0.025)	0.026 (0.030)	0.030 (0.030)	0.039 (0.030)
BMI	0.002* (0.001)	−0.006 (0.004)	0.003 (0.002)	−0.002 (0.004)	−0.002 (0.004)	−0.002 (0.004)
Income < \$2100	0.010 (0.019)	0.098 (0.090)	0.002 (0.021)	0.004 (0.022)	0.003 (0.022)	0.001 (0.022)
\$2100 ≤ Income ≤ \$5400	−0.005 (0.019)	0.006 (0.110)	−0.010 (0.018)	−0.008 (0.019)	−0.008 (0.019)	−0.009 (0.019)
\$5400 ≤ Income ≤ \$8400	−0.007 (0.019)	0.059 (0.149)	−0.007 (0.021)	−0.008 (0.021)	−0.008 (0.021)	−0.008 (0.021)
Employed	0.009 (0.015)	0.004 (0.072)	−0.016 (0.014)	−0.017 (0.014)	−0.016 (0.014)	−0.015 (0.014)
Observations	6,350	6,350	6,350	6,350	6,350	6,350
Survey cycle FE	Yes	Yes	Yes	Yes	Yes	Yes

Data source: Pooled data from the National Health and Nutrition Survey (NHANES) survey cycles 2007–2008, 2009–2010, 2013–2014, and 2015–2016. Results weighted to be nationally represented. Estimates are obtained from ordinary least square regressions weighted to account for the complex survey design. All models include indicators for survey cycles. Standard errors for ESM, Het ESM parametric, and Het ESM sieve are obtained by 1000 bootstrap replications. Stars (*0.10, **0.05, and ***0.01).

calories per day which is the USDAs dietary guidelines of minimum calories to be consumed in a day for adults who are obese (USDA 2015). Consequently, our sample consists of 6350 individuals. Since our data is survey data, all our analysis is adjusted according to the sampling weights and we include survey cycle fixed effects in all the regressions.

The outcome variable is *logkcal*, the natural log of the caloric intake reported in the dietary interview component of the survey. Our key variable of interest is *Advice*, which is a binary variable that takes the value one if the participant has been advised to lose weight by the physician. Because we allow *Advice* to be an endogenous variable, we use two instruments in our analysis: *Has health insurance* and *Has regular source of care*. The primary assumption is that because receiving physician advice is a type of medical care, the frequency of receiving this advice will be impacted by the status of health insurance and access to regular source of care. Additionally, previous research on trends in nutrient intake in the United States does not include these variables as direct determinants (King et al. 2012; Oza-Frank et al. 2009; Rehm et al. 2016). So we assume that these variables are not direct determinants of calorie consumption. In addition, research on the impact of physician advice as a health care intervention has also used variables capturing access to health insurance and routine health care as instruments (Kenkel and Terza 2001).

Other explanatory variables include demographic variables: *Age*, *Gender*, *Education*, *Employment status*, *Monthly Income*, *Marital status*, and *Race*. We also include health variables such as *BMI*, a set of binary variables indicating if the individual has (a) high blood pressure (*High BP*), (b) high cholesterol (*Cholesterol*), (c) has a coronary heart condition (*Heart condition*), (d) has diabetes (*Diabetes*), and (e) has arthritis (*Arthritis*).

All analyses were performed in Stata software (version 18, StataCorp, College Station, Texas) and accounted for NHANES sampling design and study weights. The sampling weights were normalized in accordance with NHANES guidance for using multiple cycles.

Table 2 reports the sample summary statistics of the individuals, individuals who received the physician advice to lose weight and those who did not. The average caloric consumption in our sample is about 2696 kcal. As seen from the table, only about 40.32% of the overweight individuals are advised by the physician to lose weight and individuals who are advised to lose weight have higher *BMI* on average and consume, on average, fewer calories than those who are not advised. Women, Hispanics, and individuals listed as other race are more likely to be advised to lose weight. Unsurprisingly, individuals with health conditions, such as high blood pressure, high cholesterol, a heart condition, diabetes, or arthritis are more likely to receive advice to lose weight.

5.2 | Results

In this section, we report the results from estimating endogenous switching model regressions of calorie consumption on the binary indicator of physician advice. We report the results from four estimating procedures: standard ordinary least squares (OLS), Two-stage Least Squares (2SLS), ESM with homoskedastic

treatment assignment model, Procedure 3.1, Procedure 3.2 with polynomial sieve and Procedure 3.2 with splines. For the parametric heteroskedastic probit in Procedure 3.1, we use the continuous variables *BMI*. In the second step, we specify $f_{v_0}(z_i) = f_{v_1}(z_i) = BMI$. In the Procedure 3.2 estimation with polynomials, we use power series to approximate the unknown variance functions $\sigma_{v^*}(\cdot)$, $f_{v^*_0}(\cdot)$, $f_{v^*_1}(\cdot)$. Sieve heteroskedastic probit is essentially estimated as a flexible heteroskedastic probit that includes local polynomial function of *BMI* up to two orders. In the second step, we also use local polynomial function for *BMI* up to two orders for $F_{0K_{4,N}}(z_i)$, $F_{1K_{5,N}}(z_i)$. In the Procedure 3.2 estimation with polynomials, we use splines to approximate the unknown variance functions $\sigma_{v^*}(\cdot)$, $f_{v^*_0}(\cdot)$, $f_{v^*_1}(\cdot)$. Sieve heteroskedastic probit is essentially estimated as a flexible heteroskedastic probit that includes linear splines of *BMI* with five knots. In the second step, we also use linear splines of *BMI* with five knots for $F_{0K_{4,N}}(z_i)$, $F_{1K_{5,N}}(z_i)$.

The estimated partial effects from the three probit regressions of the first stage are displayed in Table 3. Column 1 reports the results from homoskedastic probit Column 2 reports the results from heteroskedastic parametric probit estimation and Column 3 reports the results from heteroskedastic sieve probit estimation. As expected, having a health condition such as high BP, high Cholesterol, heart condition, diabetes, arthritis increases the probability of receiving the advice to lose weight. Females and Hispanics and individuals belonging to other races are more likely to receive the advice. Our excluded variables that serve as the instruments for receiving advice: *Has Health Insurance* and *Has Regular Source of Care* are individually significant as well as jointly significant in all three regressions. We also conduct a weak instrument test of Olea and Pflueger (2013) that is robust to heteroskedasticity and find the F statistic of 51.708 at the confidence level of 0.001, rejecting the null that our instruments are weak. Finally, we also report LR-statistic and Wald test statistic for testing heteroskedasticity in first stage probit. We get LR-statistic = 74.84 and Wald-statistic = 67.95 that suggest the presence of heteroskedastic in the first stage.

Table 4 reports the results from alternative specifications for calorie consumption. To assess the importance of endogeneity and heterogeneity of physician advice, column (1) reports the results from a simple OLS specification. Column (2) report the results from a 2SLS regression where *Advice* is instrumented by health coverage and having a regular source of care. Both the specifications also include interaction variables where the demeaned covariates are interacted with the treatment (physician advised to lose weight). We find a negative and statistically significant result of receiving physician advice to lose weight on calorie consumption: on average, overweight individuals who are advised to lose weight reduce their predicted calorie consumption by about 1%. However, 2SLS estimates for *Advice* does not significantly effect predicted calorie consumption. In fact, 2SLS estimates for physician advised to lose weight are positive but statistically insignificant. This unrealistically positive and large estimate is certainly difficult to comprehend. However, it does suggest that we should use caution while applying basic IV methods when substantial heterogeneity may be present.

The results from the standard Endogenous Switching Model (ESM) reported in column (3), where we only account for the

endogeneity of physician advice show that the physician advice does lead to a reduction in the average predicted consumption by about 8%, but this effect is not statistically significant. Column (4) reports the results from parametric heteroskedastic ESM with constant coefficient,s and column (5) reports the results from sieve heteroskedastic ESM with polynomial sieves. Finally, Column (6) reports the results from sieve heteroskedastic ESM with splines sieves. We see that physician advice to lose weight results in about 14% decrease in average predicted calorie consumption in both parametric and sieve estimation and is found to be significant. Our results highlight the importance of accounting for heterogeneity in treatment assignment as captured by heteroskedasticity.

6 | Conclusion

We have provided parametric and semiparametric estimation procedures for endogenous switching models that allow the treatment variable to follow a heteroskedastic probit model with a flexible heteroskedastic function and substantial heterogeneity in the relationship between the unobservables in the potential outcome equations and the treatment equation. The key to our approach is using conditional linear projections, which leads to estimating equations. From there, we can use a parametric approach or a semiparametric approach based on the method of sieves. In addition to allowing but separate coefficients in the two potential outcome equations, we consider the case where the coefficients on the control variables can be depend on unobserved heterogeneity.

We use our methodology to estimate the effect of receiving physician advice to lose weight on calorie consumption, we find that when we account for heteroskedasticity in the errors of the treatment assignment, physician advice has a practically substantial and significant impact on calorie consumption. In contrast, the traditional least squares, instrumental variable estimation methods that ignore the heterogeneity in the treatment assignment fail to capture this effect. Our findings suggest the importance of physician advice as a low-cost intervention to prevent rising obesity and related health issues.

There are several ways in which we could extend the model and methodology of our paper. In the [Supporting Information: Appendix C](#), we provide a preliminary extension of our methodology to heterogeneous coefficients ESMs. We would like to allow some covariates to be endogenous in the outcome equation. We would also like to consider panel data endogenous switching models with unobserved effects. We leave such extensions to future research.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data have been provided to the journal, and it is available at *Journal of Applied Econometrics Data Archive*.

OPEN RESEARCH BADGES



This article has been awarded Open Data Badge for making publicly available the digitally-shareable data necessary to reproduce the reported results. Data is available at <https://doi.org/10.15456/jae.2025087.2159666707>.

Endnotes

¹ These corrections functions are obtained as generalized residuals in the form of $w_i[\lambda(z_i\gamma)] - (1 - w_i)[\lambda(-z_i\gamma)]$ where $\lambda(\cdot)$ is the Inverse Mills Ratio.

² Throughout, we assume random sampling from the population, in which case we can state assumptions for a single random draw.

³ Assumption 2.3(a) formalizes the notion of exogeneity of the instruments. Specifically, only the treatment indicator w_i is allowed to be endogenous in the sense that only w_i is allowed to be correlated with u_i^0 and u_i^1 . Assumption 2.3(b) imposes an exclusion restriction for identification. Specifically, we assume that we have at least one variable that exclusively affects the treatment mechanism and does not also impact the outcome variable. Though technically, we can still identify the parameters when $K_2 = 0$, this exclusion restriction is imposed to avoid achieving identification only due to the nonlinearities of the control functions (Wooldridge 2010).

⁴ With respect to Assumption 2.4, it is important to understand that z_i cannot appear as an additive function in specifying CLPs. For example, we cannot specify $E[u_{ij}|v_i, z_i] = \rho_{j0} + \rho_{j1}v_i + z_i\rho_{j2}$ for $j = 0, 1$ because then the parameters in the potential outcome equations will not be identified. See Section 3.3 for details.

⁵ In control function approaches without heteroskedasticity these functions are assumed to be constant.

⁶ The simplest specification is a linear one with $F_{v_0}(z_i) = F_{v_1}(z_i) = z_i$.

⁷ In practice, this would imply running the standard *hetprobit* command in Stata.

⁸ Alternatively, we can also run a bootstrap routine.

⁹ As pointed out by Lewbel (2019), in this case, point identification will fail. The model will be set identified, the identified set consisting of the possible solutions.

¹⁰ We thank an anonymous referee for pointing this out.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.